

COMP 693 Industry Project

Final Report

Azure Hybrid Cloud Solution Design and Implementation

(Independent Project)

Submitted

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EXECUTIVE SUMMARY

For this project, I created a hypothetical company called KR Industries. As time passed and the business grew, the company's IT infrastructure could no longer meet the increasing demands for computing efficiency. However, the cost of investing in new physical devices was very high, and maintaining the system was placing a heavy burden on the IT team. Therefore, the company decided to use Azure cloud computing to implement a hybrid cloud migration plan, retaining control of some on-premises resources while gradually moved other applications to the Azure public cloud. This plan not only reduced the company's dependence on local hardware and lowered operating costs but also increased flexibility and scalability. It enabled the company to adjust resource allocation based on business needs, which improved management efficiency. Despite some resource limitations, the successful migration of critical applications demonstrated the clear benefits of cloud migration.

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GLOSSARY/ACRONYMS

- **Azure:** Microsoft's cloud service that provides many tools like computing, storage, analytics, and networking.
- **Hybrid Cloud:** A setup that mixes private data centers with public cloud services to work together.
- **TCO (Total Cost of Ownership):** A tool to figure out the total cost of owning and managing resources over time.
- **VM (Virtual Machine):** A virtual computer that can run programs and operating systems remotely. **SSH (Secure Shell):** A protocol used to access and manage remote computers and databases securely.
- **DMS (Data Migration Service):** A service that helps move data from local databases to Azure. It was used to migrate the MySQL database in this project.
- **NSG (Network Security Group):** A set of rules in Azure to control incoming and outgoing traffic for virtual machines.
- **MySQL Workbench:** A tool used to manage, design, and work with MySQL databases.
- **GitHub Actions:** A tool in GitHub that automates tasks like deploying applications, such as web apps to Azure.
- **Azure App Service Plan:** A resource in Azure that runs web apps, APIs, and mobile app backends.
- **Azure Blob Storage:** A cloud service for storing large files like images, videos, and backups.
- **Recovery Services Vault:** A service that helps manage backups and recovery for virtual machines in Azure.
- **Azure Key Vault:** A service in Azure for storing and managing passwords, API keys, and certificates securely.
- **Microsoft Defender for Cloud:** A security service in Azure that helps protect cloud resources by detecting threats and giving recommendations to improve security.
- **Azure Advisor:** A service that gives tips to improve the performance, security, and cost of Azure resources.
- **Azure Backup:** A service for backing up and restoring virtual machines and other resources in Azure.

1. BACKGROUND

1.1 Overview

KR Industries is a traditional medium-sized manufacturing company located in Wellington, New Zealand, which mainly produces mechanical parts. The company currently has about 400 employees, and the company structure includes four main departments: production, sales, finance, and IT. As the business grows, KR Industries is facing challenges to keep its current IT infrastructure updated. The existing system is not handling the increasing demands well, which causes higher operational costs and inefficiencies. The company needs a more flexible and scalable solution to support production needs and lower operational risks to stay competitive.

1.2 Problem

KR Industries mainly faced challenges with costs and resource allocation.

- The existing IT infrastructure could not meet the growing business needs, and adding more physical devices was not only expensive but also required additional maintenance costs.
- The company could not flexibly adapt to seasonal production changes. During peak times, the current infrastructure could not expand fast enough to meet growing demand, while in non-peak times, keeping the same resources led to inefficiency and waste.
- There were some other risks, such as security problems or hardware failures, which could seriously impact operations and even cause business interruptions.

1.3 Project Team

I am the only person for this project

2. REQUIREMENTS AND GOALS

2.1 Overall goal

The main goal of this project is to help KR Industries address its challenges by reducing IT costs, optimizing resource use, and improving work efficiency and security. This will be done by creating and implementing a hybrid cloud migration plan, which will move critical applications to the Public cloud.

2.1.2 Sub-goals

- Reduce the reliance on physical infrastructure
- Optimize resource allocation by implementing scalable cloud services.
- Migrate critical applications to the cloud
- Improve security
- Support remote work

2.2 Requirement

- Design a hybrid cloud migration architecture that balances on-premise and public cloud workloads.
- Set up necessary cloud resources for the migration.
- Migrate critical applications to the cloud and create basic backup plans.
- Set up security measures to protect company data and systems.
- Provide safe remote management through public IP addresses and SSH access.
-

2.2.1 How the Requirements are Met

- The architecture diagram shows the configuration of various resources and the migration process from on-premise to the cloud.
- Cloud resources (e.g., virtual machines, databases, virtual networks, and storage) are set up to support flexible and efficient resource management. Virtual machines can be stopped during off-peak times to reduce costs.
- Critical applications such as SQL databases and web applications are successfully migrated to the cloud, and backup plans ensure the protection of essential business data.
- Basic security services are set up correctly
- Virtual machines (VMs) can be safely managed through SSH access, allowing for remote work such as changing data and running processes.

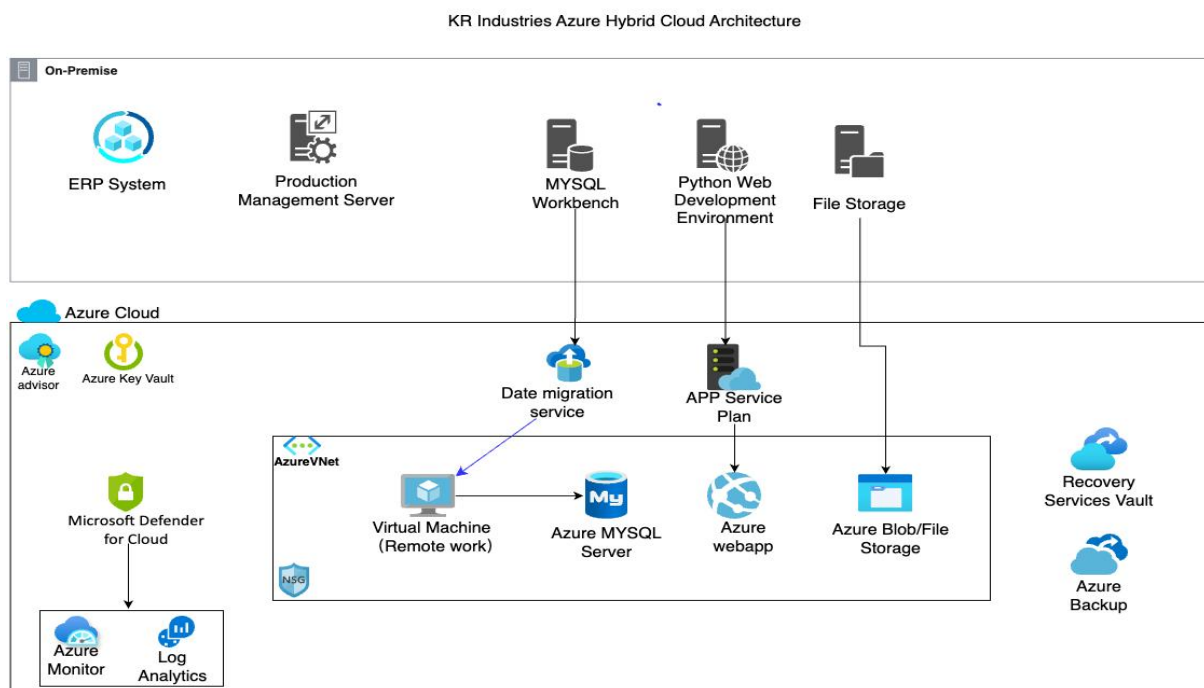
3. METHODS

3.1 Overview

This project followed several key steps and used various tools to achieve a successful hybrid cloud migration for KR Industries. Each of these tools was important in helping the project achieve its goals and ensuring a smooth migration process.

- **TCO (Total Cost of Ownership):** I started by estimating the cost of moving to the cloud. This helped me understand how much the company could save on IT costs.
- **Draw.io:** I used Draw.io to design the architecture diagram for the hybrid cloud. This helped me make sure all resources were well organized, which supports the goal of using resources better.
- **Azure Portal:** Azure Portal was the primary platform for setting up and managing cloud resources. I set up all the necessary cloud resources, like virtual machines and databases. This tool helped make sure that the company's resources could grow as needed.
- **GitHub :** I used GitHub to store the code for the web app, and GitHub Actions to deploy it to Azure. This helped to make the deployment process easier and more efficient.
- **MySQL Workbench:** This was used to move the company's databases to the cloud. It helped ensure that essential data was successfully migrated, which supports both efficiency and security goals.
- **Python:** I used Python to create a simple web app, and together with GitHub, it helped demonstrate how cloud-based application management works.

3.2 Design



For this project, I created a simplified hypothetical company environment that represents some critical systems of KR Industries, focusing on their IT infrastructure. The architecture reflects a hybrid cloud approach, where some systems remain on-premise while others are migrated to the cloud.

The architecture has two main sections: on-premise and Azure Cloud:

3.2.1 On-premise

- **ERP System and Production Management Server:** These are included as placeholders to represent real-world systems that might still be running on-premise in a company. They show possible future migration opportunities.
- **MySQL Workbench:** I used MySQL Workbench to connect to and manage the Azure MySQL Database. This was done through an SSH tunnel created on the Azure's Virtual Machine.
- **Python Web Development Environment:** I used Python to develop a basic web application for the project. After that, I deployed it to Azure Web Apps through GitHub and Azure App Service Plan.

3.2.2 Azure cloud environment

- **Virtual Machine (Remote work):** I used this VM to establish an SSH tunnel, which was necessary to connect MySQL Workbench to the Azure MySQL server. The VM also allows for secure remote work by modifying data or running process directly on the server.
- **Azure MySQL Database:** The company's database was migrated to Azure's MySQL Database using Azure's Data Migration Service (DMS). I had to set up MySQL Workbench with the right connection details, like the server's public IP address, SSH information, and login details.
-
- **Azure Web App:** I developed a simple Python web application as an example to demonstrate cloud migration. I built this app on Python first, then push it to GitHub. Then, I used Azure's App Service Plan to put it on Azure's web server. This showed how GitHub Actions can work with Azure for continuous deployment.
- **Azure Blob/File Storage:** For the file migration part, I created some essential files, such as Excel sheets and images, which were moved to Azure Blob Storage. This part simulates how the company can store and manage large data files securely and flexibly in the cloud.
- **Azure Security Measures:** Security was another important part of the project. I setup basic security settings, including **Network Security Groups (NSG)** to control inbound and outbound traffic, **Azure Key Vault** to manage sensitive information like passwords, **Azure Advisor** to provide recommendations for improving security and cost management, and **Microsoft Defender for Cloud** to monitor and protect cloud resources.

- **Backup and Recovery:** I created a Recovery Services Vault for backup, but because of the limitations with the VM configuration (trusted launch), I could not fully use Azure Site Recovery. However, the basic backup service is working to protect critical data.

Overall, this design helps achieve the project's sub-goals by ensuring critical workloads are securely migrated to the cloud, while maintaining necessary systems on-premise to support the hybrid cloud strategy.

3.3 Risks and Challenges

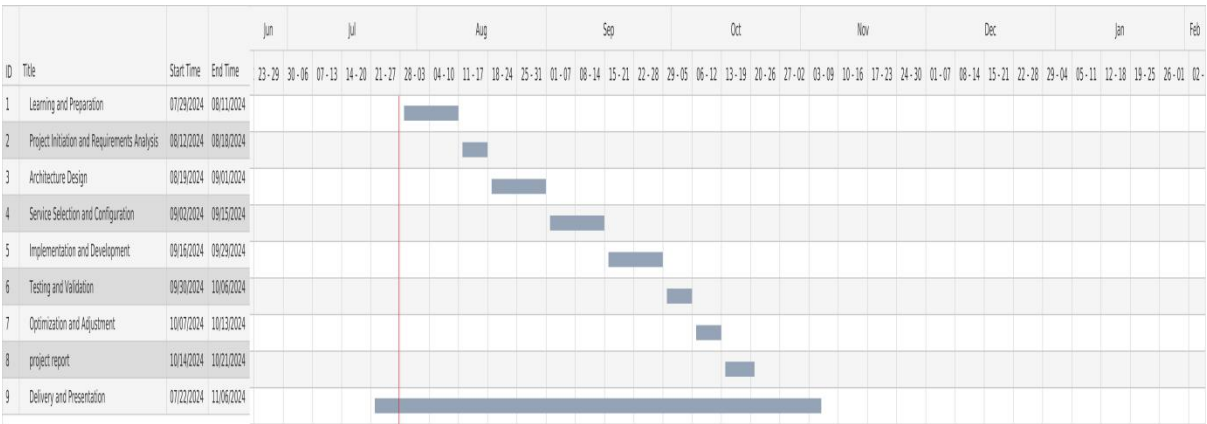
During this project, I faced several risks and challenges.

- **Resource limitation:** I used a student Azure subscription, which has a 100 dollar quota limitation. Therefore, I could not use many advanced features like high-performance virtual machines. Security was another limitation. I could only use basic security measures, like network security groups, which limited the scale and depth of the cloud migration I could demonstrate. However, it's important to note that not all companies need the highest level of VMs or security configurations - it depends on what the business needs. Otherwise, they might waste resources and increase their cost. To deal with this limitation, I focused on using the basic services in my subscription. I made the design simpler but kept the main parts needed to show the migration process. I also used security measures that fit the size of my demonstration.
- **Simulate a real company environment:** Another challenge was making it like a real company environment. Since this was a hypothetical company, I could not make it as complex as a real one. Real companies have more data and larger volumes of files, and their web applications have more features - like handling real-time traffic or processing orders. My simple web app does not have those functions. However, I did manage to deploy my Python web app on Azure WebApp using GitHub Actions. This showed how to move an application to the cloud. I also set up an SSH tunnel to connect to and manage the Azure MySQL database. This helped keep the data migration safe and easy to manage. I tried to create virtual applications and services to show how to move a basic setup to the cloud.
- **Disaster recovery:** My virtual machines used Trusted Launch, which meant I couldn't use Azure Site Recovery. I also couldn't choose multiple regions or set up geographic backup. These would have been useful for testing a full disaster recovery plan. Because of these limits, I could not fully show the type of disaster protection a real company would need. I have to keep backup recovery for this. However, I did set up Azure Backup as part of the recovery plan, ensuring that critical data is regularly backed up and recoverable.

Despite these challenges, I finished the migration and showed how a basic hybrid cloud setup works. While the project could have been bigger with more resources, I worked within the limitations and showed how moving to the cloud can make a company more flexible, secure, and cost-efficient.

3.4 Implementation

This is my project Gantt chart, it clearly shows the timeline of the whole project and the progress of each stage. I will detailed explain the steps of the project implementation below:



- **Learning and Preparation**

Since cloud computing was new to me, I spent two weeks learning the basics of Microsoft Azure. During this time, I became familiar with the main tools and concepts on the platform. This learning prepared me for the architecture design and cloud migration steps later on. I also successfully signed up for Azure and learned how to use the Azure portal effectively.

Tools: Microsoft Learn, Azure documentation and videos

Artifacts: Azure Learning.docx, where I noted my learning process.

- **Project initiation and requirements analysis**

In this phase, I created the IT structure for the hypothetical company KR Industries and did a Total Cost of Ownership (TCO) analysis to estimate the costs of moving from on-premises infrastructure to Azure. The estimate was based on a real company environment, with assumptions for virtual machines, storage, and database services. Due to resource limits, the final setup is not the same as the original estimate. However, this analysis is still useful for future cloud migration and expansion. These helped guide the project steps, ensuring it would meet goals like cost optimization and better resource usage. (The TCO report is shown in Appendix D, Figure 1.)

Tool: TCO calculator

Artifacts: KR Industries outline.docx (company background and needs), TCO document (initial cost estimates)

- **Architecture Design**

Based on the needs analysis, I designed the hybrid cloud architecture and used Draw.io to make the architecture diagram. This design made sure workloads could be balanced between on-premise systems and the cloud. It gave a clear framework for setting up services and moving applications later.

Tool: Draw.io

Artifacts: KR Hybrid Cloud Architecture Diagram and Hybrid Cloud Architecture Draft

- **Service Selection and Configuration**

In this phase, I set up the necessary Azure services based on the architecture design. This included setting up virtual machines, MySQL databases, Blob storage, and security services like NSG and Key Vault. After setting everything up, I did some tests to make sure these services met the project needs and were cost-efficient.

Tool: Azure portal

Artifacts: Azure service configuration and testing.docx, recording the setup and testing.

- **Implementation and Development**

This phase focused on executing the project. I migrated the local MySQL database to Azure using MySQL Workbench and an SSH tunnel. I also created a simple Python web application locally and deployed it to Azure WebApp using GitHub Actions. I also set up file storage and backups.

Tools: MySQL Workbench, Python development environment, GitHub Actions, Azure WebApp

Artifacts: KRIndustries Webapp URL, GitHub repository

- **Testing and Verification**

I used Azure Monitor and Log Analytics to monitor system performance and ensure everything worked as expected. I also tested backup and recovery strategies to confirm data reliability and ensure the business could continue running even in case of issues.

Tools: Azure Monitor, Azure Backup

Artifacts: Azure Monitor logs, showing the performance tests and backup verification.

- **Optimization and Adjustment**

After testing, I made some adjustments to the system. This included adjusting the virtual machine settings and security configurations to make sure the system stayed cost-effective and used resources efficiently.

Tools: Azure portal, NSG, Azure Monitor

Artifacts: project summary.docx

4. RESULTS AND OUTCOMES

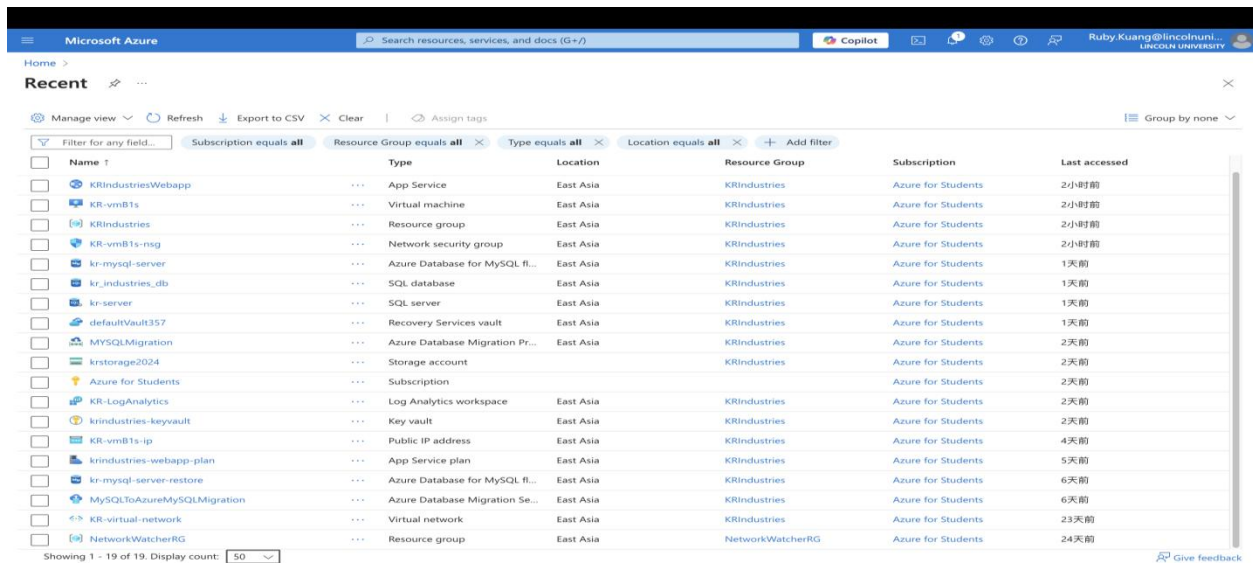
4.1 Evidence of Deliverables

To demonstrate how the project met its overall goals and requirements, I have included evidence of the system configuration, migration, and deployment processes that were crucial to achieving a successful hybrid cloud migration for KR Industries. Additional evidence can be found in the Appendix.

4.1.1 Azure services Configuration

All the critical resources I configured for the project are organized under the KRIndustries resource group. This setup was essential for the project, as it kept all the cloud services in one place for easier management.

Evidence: In the screenshots provided, you can see the Azure portal with all the critical resources created, including: Virtual Machine (KR-vmB1s), Azure MySQL server (kr-mysql-server), Web app service (KRIndustriesWebapp), Storage account (krstorage2024)



The screenshot displays the Microsoft Azure portal interface. At the top, there's a navigation bar with the Microsoft Azure logo, a search bar, and user information for Ruby Kuang. Below the navigation bar, the 'Recent' view is active, showing a list of resources. The resources are filtered by 'Subscription equals all', 'Resource Group equals all', 'Type equals all', and 'Location equals all'. The table lists various resources including KRIndustriesWebapp, KR-vmB1s, KRIndustries, KR-vmB1s-nsg, kr-mysql-server, kr_industries_db, kr-server, defaultVault357, MySQLMigration, krstorage2024, Azure for Students, KR-LogAnalytics, krindustries-keyvault, KR-vmB1s-ip, krindustries-webapp-plan, kr-mysql-server-restore, MySQLToAzureMySQLMigration, KR-virtual-network, and NetworkWatcherRG. Each row includes a checkbox, name, type, location, resource group, subscription, and last accessed time.

Name	Type	Location	Resource Group	Subscription	Last accessed
KRIndustriesWebapp	App Service	East Asia	KRIndustries	Azure for Students	2小时前
KR-vmB1s	Virtual machine	East Asia	KRIndustries	Azure for Students	2小时前
KRIndustries	Resource group	East Asia	KRIndustries	Azure for Students	2小时前
KR-vmB1s-nsg	Network security group	East Asia	KRIndustries	Azure for Students	2小时前
kr-mysql-server	Azure Database for MySQL flex	East Asia	KRIndustries	Azure for Students	1天前
kr_industries_db	SQL database	East Asia	KRIndustries	Azure for Students	1天前
kr-server	SQL server	East Asia	KRIndustries	Azure for Students	1天前
defaultVault357	Recovery Services vault	East Asia	KRIndustries	Azure for Students	1天前
MySQLMigration	Azure Database Migration Project	East Asia	KRIndustries	Azure for Students	2天前
krstorage2024	Storage account	East Asia	KRIndustries	Azure for Students	2天前
Azure for Students	Subscription			Azure for Students	2天前
KR-LogAnalytics	Log Analytics workspace	East Asia	KRIndustries	Azure for Students	2天前
krindustries-keyvault	Key vault	East Asia	KRIndustries	Azure for Students	2天前
KR-vmB1s-ip	Public IP address	East Asia	KRIndustries	Azure for Students	4天前
krindustries-webapp-plan	App Service plan	East Asia	KRIndustries	Azure for Students	5天前
kr-mysql-server-restore	Azure Database for MySQL flex	East Asia	KRIndustries	Azure for Students	6天前
MySQLToAzureMySQLMigration	Azure Database Migration Project	East Asia	KRIndustries	Azure for Students	6天前
KR-virtual-network	Virtual network	East Asia	KRIndustries	Azure for Students	23天前
NetworkWatcherRG	Resource group	East Asia	NetworkWatcherRG	Azure for Students	24天前

4.1.2 Virtual Machine Setup and Remote Work

I set up a virtual machine and used an SSH connection to work remotely on it. The VM was used to create an SSH tunnel for the database migration, and it could also be accessed via a public IP address: <http://20.205.21.46/>

Evidence: The screenshot shows the successful SSH connection to the VM. This is an important step because it proves the remote access to the system is working, allowing me to securely manage the VM from anywhere. This is part of the project's sub-goal to support remote work and flexible management.

```
isaack — Ruby@KR-vmB1s: ~ — ssh Ruby@20.205.21.46 — 163x63

To update your account to use zsh, please run `chsh -s /bin/zsh`.
For more details, please visit https://support.apple.com/kb/HT208050.
MacBook-Air-71~ isaack$ ssh Ruby@20.205.21.46
Ruby@20.205.21.46's password:
Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 6.8.0-1015-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Tue Oct  8 23:13:17 UTC 2024

System load:  0.0               Processes:    121
Usage of /:   9.7% of 28.02GB   Users logged in: 0
Memory usage: 78%              IPv4 address for eth0: 10.1.0.4
Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
just raised the bar for easy, resilient and secure K8s cluster deployment.
https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

14 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Tue Oct  8 22:22:59 2024 from 202.160.61.122
Ruby@KR-vmB1s:~$ mysql -u root -p
Enter password:
ERROR 1045 (20000): Access denied for user 'root'@'localhost' (using password: YES)
Ruby@KR-vmB1s:~$ mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 9
Server version: 8.0.39-0ubuntu0.24.04.2 (Ubuntu)

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

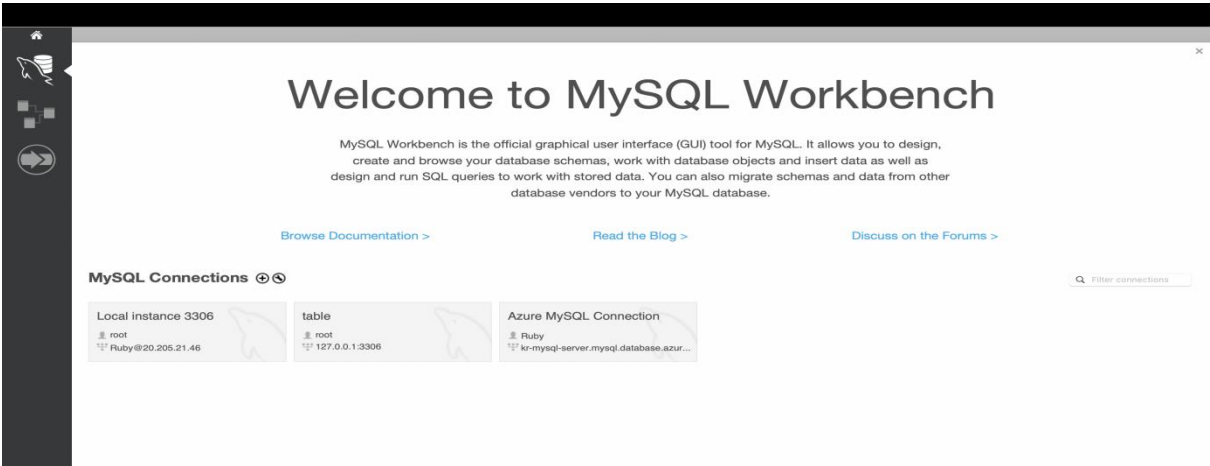
mysql> show databases;
+-----+
| Database |
+-----+
| information_schema |
| kr_industries |
| mysql |
| performance_schema |
| sys |
| testdb |
+-----+
6 rows in set (0.03 sec)

mysql>
```

4.1.3 SQL Database Migration

One key goal of this project was to migrate the company’s MySQL database to Azure. I used MySQL Workbench to do this, with an SSH tunnel on the virtual machine to ensure secure data transfer.

Evidence: The first screenshot shows MySQL Workbench successfully connected to the Azure MySQL database. This confirms that the SSH connection was made securely. For more details on the SSH tunnel configuration, see Appendix A, Figure 2.



The next screenshot shows the successful migration of key tables (Customers, Inventory, Orders, etc.) to the Azure database was carried out using Data Migration Service (DMS). This proves the data was migrated correctly to Azure database. DMS setup can be found in Appendix A, Figure 3.

Name	Status	Migration details	Duration
KRIndustriesDB.Customers	Completed	All rows completed (~10).	00:00:00
KRIndustriesDB.Inventory	Completed	All rows completed (~10).	00:00:00
KRIndustriesDB.Orders	Completed	All rows completed (~10).	00:00:00
KRIndustriesDB.ProductionTasks	Completed	All rows completed (~10).	00:00:00
KRIndustriesDB.Products	Completed	All rows completed (~10).	00:00:00

4.1.4 Azure WebApp Deployment

For the web application component of this project, I deployed a simple Python-based web app to Azure WebApp using GitHub Actions. The screenshot below (KRIndustriesWebapp Overview) provides evidence of the successful deployment and setup, showing the app running in the Azure environment.

Evidence:

- The web application URL: <https://krindustrieswebapp-h3aggbg4d8gmhbeb.eastasia-01.azurewebsites.net/>
- GitHub Project: <https://github.com/RubyKuang/Azure-HybridCloudMigration>

(A simple webapp was created to demonstrate how to migrate a web application to Azure only.) For more detailed evidence, including screenshots of the Github Action pages, please refer to Appendix A, Figure1.

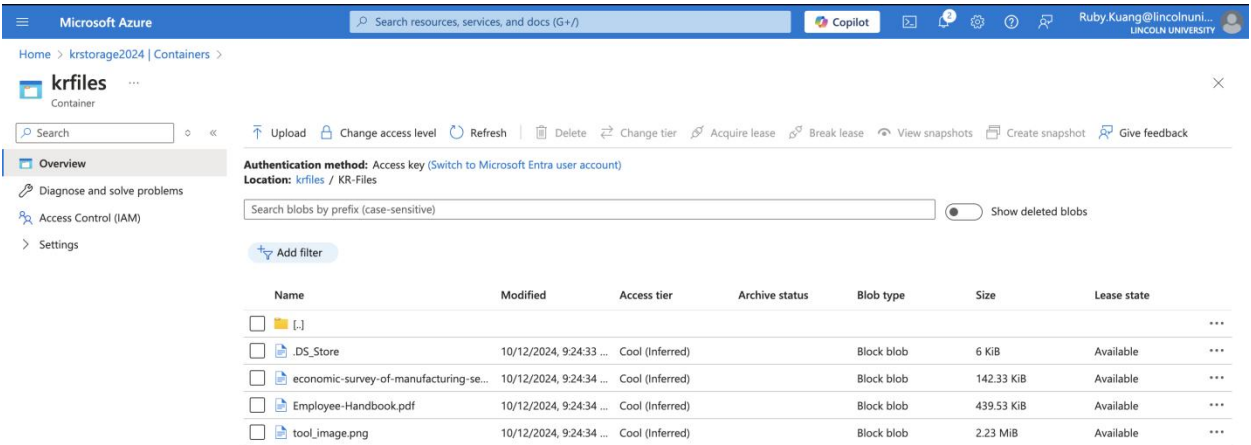
- KRIndustriesWebapp Overview

Property	Value
Resource group	KRIndustries
Status	Running
Location	East Asia
Subscription	Azure for Students
Subscription ID	12b90514-ee41-4f67-b1da-d553ce72ed6b
Tags	AppName : WebServer
Default domain	krindustrieswebapp-h3aggbg4d8gmhbeb.eastasia-01.azurewebsites.net
App Service Plan	krindustries-webapp-plan (F1: 1)
Operating System	Linux
Health Check	Not Configured
GitHub Project	https://github.com/RubyKuang/Azure-HybridCloudMigration
Name	KRIndustriesWebapp
Publishing model	Code
Runtime Stack	Python - 3.11
Default domain	krindustrieswebapp-h3aggbg4d8gmhbeb.eastasia-01.azurewebsites.net
Custom domain	Add custom domain
Plan Type	App Service plan
Name	krindustries-webapp-plan
Operating System	Linux
Instance Count	1
SKU and size	Free (F1) Scale up

4.1.5 Azure File Storage Migration

In this part of the project, I migrated some important files to Azure Blob Storage. This step was important to make sure company documents are safe, which was part of the goal for storing and protecting important data in the cloud.

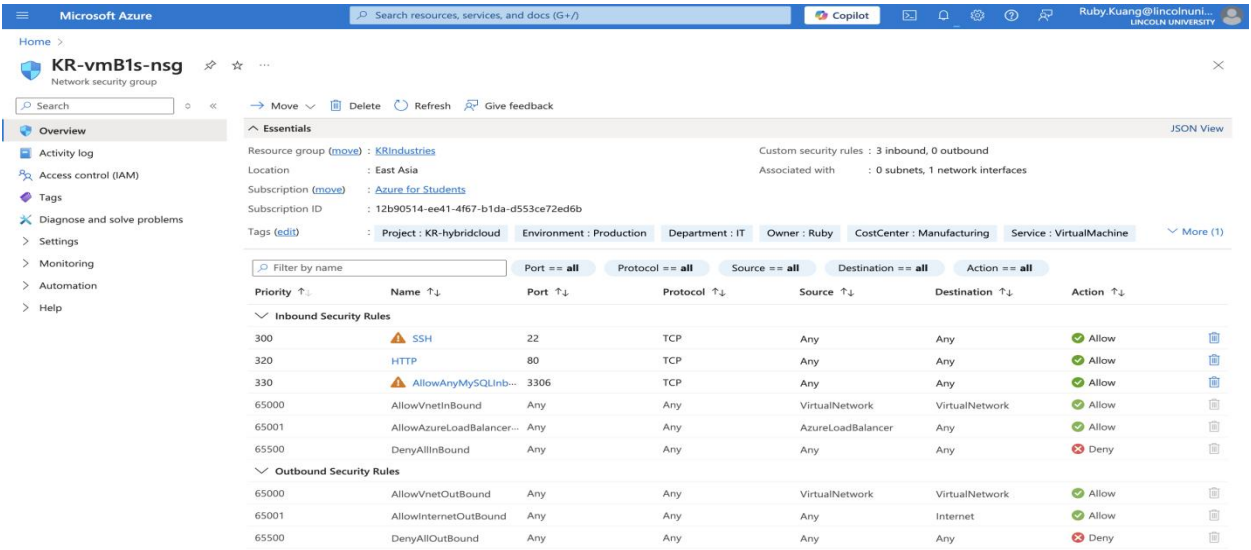
Evidence: The screenshot shows files like PDFs and an image uploaded to the container named “krfiles” inside the storage account “krstorage2024.” This proves that I successfully moved files to the cloud for storage and backup.



4.1.6 Security Configuration

I set up some basic rules using the Network Security Group (NSG) to keep the virtual machines and applications secure. This way, I made sure only the needed connections could reach the virtual machine, reducing the chances of security risks.

Evidence: The screenshot below shows the rules I used to control inbound and outbound traffic. For example, I allowed access to SSH, HTTP, and MySQL while blocking other unnecessary traffic.

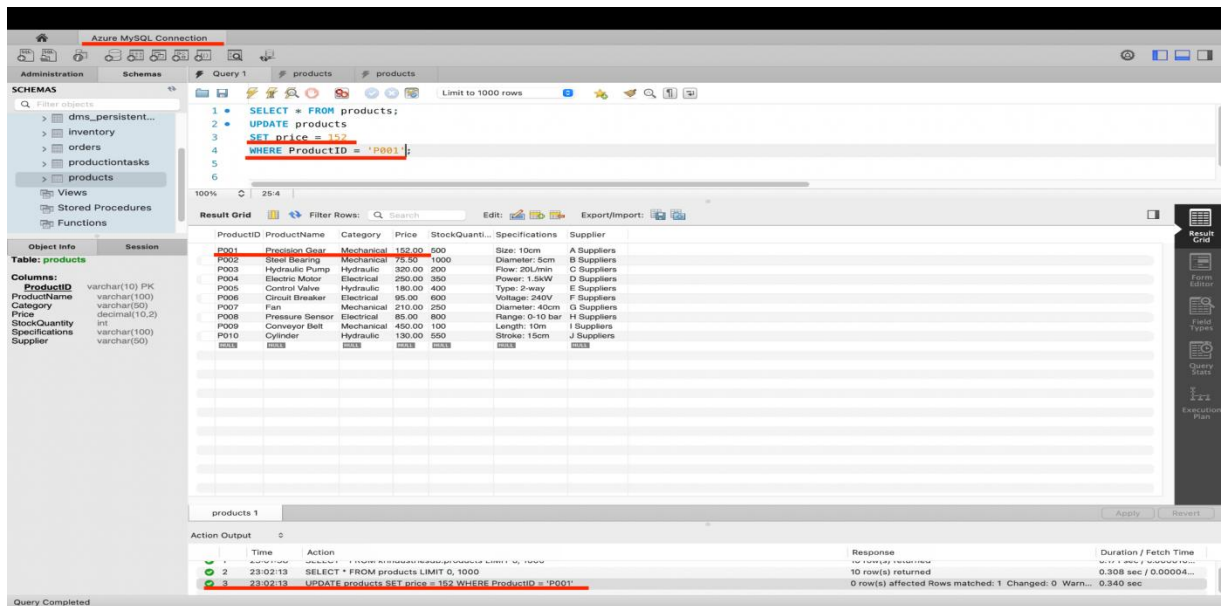


4.2 Testing/Validation

4.2.1 Functional Testing

I tested the basic functions to make sure the database and web app worked after migration. These tests showed that they worked properly.

- **Database:** I connected to the Azure MySQL database using MySQL Workbench. I updated the price of product P001 from 150.00 to 152.00. The update worked, and I checked the database to confirm the change.



- **Web Application:** After I updated the database, I checked the Azure web app to see if the new price shows up. The products page showed the updated price for P001, so I know the web app is displaying the correct data.

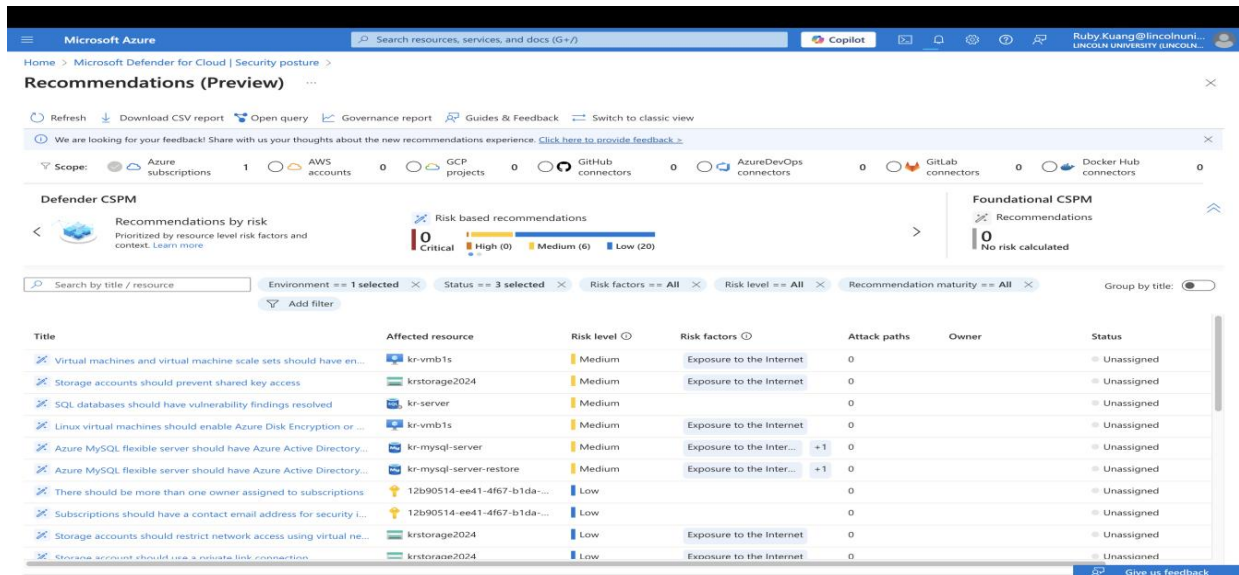


Products

Product ID	Product Name	Category	Price	Stock Quantity	Specifications	Supplier
P001	Precision Gear	Mechanical	152.00	500	Size: 10cm	A Suppliers
P002	Steel Bearing	Mechanical	75.50	1000	Diameter: 5cm	B Suppliers
P003	Hydraulic Pump	Hydraulic	320.00	200	Flow: 20L/min	C Suppliers
P004	Electric Motor	Electrical	250.00	350	Power: 1.5kW	D Suppliers
P005	Control Valve	Hydraulic	180.00	400	Type: 2-way	E Suppliers
P006	Circuit Breaker	Electrical	95.00	600	Voltage: 240V	F Suppliers
P007	Fan	Mechanical	210.00	250	Diameter: 40cm	G Suppliers
P008	Pressure Sensor	Electrical	85.00	800	Range: 0-10 bar	H Suppliers
P009	Conveyor Belt	Mechanical	450.00	100	Length: 10m	I Suppliers
P010	Cylinder	Hydraulic	130.00	550	Stroke: 15cm	J Suppliers

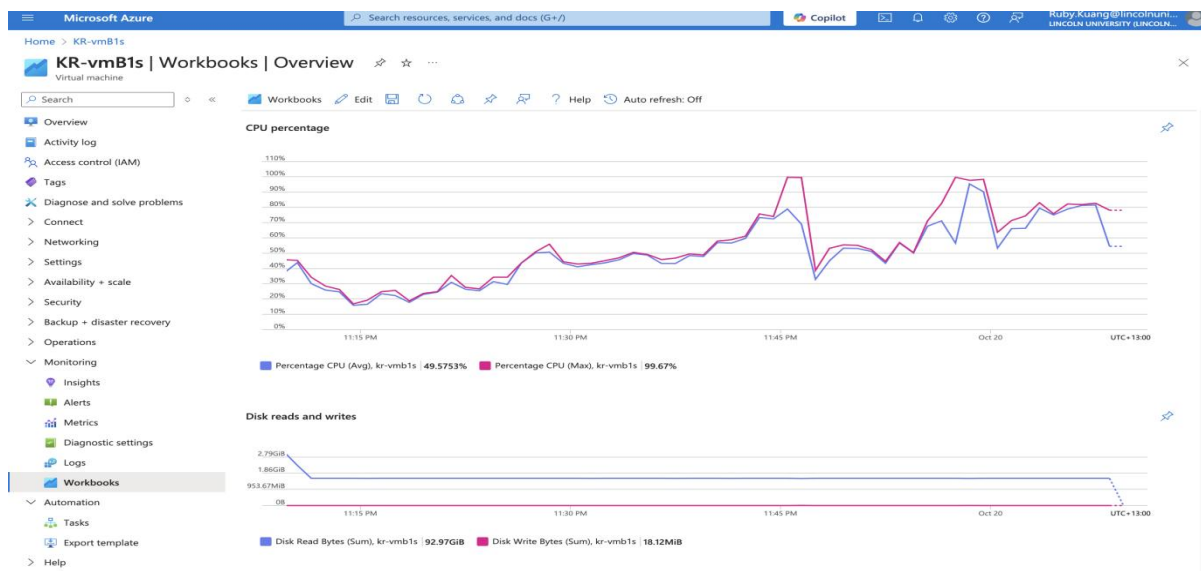
4.2.2 Security Testing

I used Azure Defender to check the security of my resources. It gave me recommendations to improve security and showed risks for each resource. I checked the settings based on these recommendations and made changes where needed. This helped reduce the risk of exposing resources to the internet. For additional security recommendations from Azure advisor, see the Appendix C, Figure 1.



4.2.3 Performance Testing

During the performance testing phase, I monitored several metrics, including CPU, disk, network, and memory usage. While testing the virtual machine, I found that the CPU usage peaked at 99%, which caused connection problems. As a result, I resized the VM from B1 to B2 for more resources to handle the workload better. The screenshot below shows the CPU and disk usage for the VM. Other metrics for different applications, can be found in the Appendix B.



4.2.4 Backup testing

I set up regular backups for the virtual machine using Azure Backup. As the screenshot shows, several recovery points have been created, including crash-consistent and file-system-consistent snapshots. These backups make sure the system can be restored if needed. The restore points of the virtual machine can be found in Appendix C, Figure 2.

The screenshot displays the Azure portal interface for the 'KR-vmB1s' virtual machine backup. The left sidebar shows navigation options like Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, and Backup + disaster recovery. The main content area shows the backup configuration, including the subscription ID, backup policy, alerts, and jobs. Below this, a table lists the recovery points, categorized by consistency type: Crash Consistent (31), Application Consistent (0), and File-System Consistent (17). The table columns are Creation time, Consistency, Recovery type, and Expiry time.

Creation time	Consistency	Recovery type	Expiry time
10/20/2024, 1:41:18 AM	File-system Consistent	Snapshot	...
10/19/2024, 9:10:44 PM	Crash Consistent	Snapshot	...
10/19/2024, 9:08:12 AM	Crash Consistent	Snapshot and Vault-Standard	...
10/19/2024, 5:21:07 AM	Crash Consistent	Snapshot	10/25/2024, 4:21:07 PM
10/19/2024, 1:12:48 AM	Crash Consistent	Snapshot	10/25/2024, 12:12:48 PM
10/18/2024, 10:22:59 PM	Crash Consistent	Snapshot	10/25/2024, 9:22:59 AM
10/18/2024, 9:08:21 AM	Crash Consistent	Snapshot and Vault-Standard	10/24/2024, 8:08:21 PM
10/18/2024, 5:15:09 AM	Crash Consistent	Snapshot	10/24/2024, 4:15:09 PM
10/18/2024, 1:13:53 AM	Crash Consistent	Snapshot	10/24/2024, 12:13:53 PM
10/17/2024, 9:11:00 PM	Crash Consistent	Snapshot	10/24/2024, 8:11:00 AM
10/17/2024, 9:06:05 AM	Crash Consistent	Snapshot and Vault-Standard	10/23/2024, 8:06:05 PM
10/17/2024, 5:15:40 AM	Crash Consistent	Snapshot	10/23/2024, 4:15:40 PM
10/17/2024, 1:09:54 AM	Crash Consistent	Snapshot	10/23/2024, 12:09:54 PM

5. REFLECTIONS

5.1 Reflections

Overall, I achieved the main goal of this project. I managed to build a hybrid cloud architecture and migrated several key applications from local servers to the public cloud based on the architecture design. Even though I faced a lot of issues with resource configuration because of the student subscription limits, I was still able to complete the core parts of the project.

New learning:

- **Basic cloud computing concepts:** I had to dive deep into the basics of Azure and cloud computing to understand how cloud migration works and its principles.
- **Architecture design:** I learned how to design a hybrid cloud architecture and figure out which services should be migrated to the cloud.
- **Resource configuration:** I had to understand how to configure different resources in Azure, like virtual machines, storage, and network security groups (NSG), and learn what each one is used for.
- **SSH tunnel setup:** For the database migration, I needed to learn how to set up an SSH tunnel to connect my local MySQL Workbench to the cloud database.
- **GitHub Actions for automation:** I also learned how to automate the deployment of my webapp using GitHub Actions, which was very useful for my project's automation.

Personal growth:

- **Patience:** The project was more complex than I expected, with many configurations requiring repeated adjustments and tests. I learned how to stay patient and persistent when faced with difficulties.
- **Time management:** The initial project proposal plan, the milestones, and the weekly journals/review meetings helped me understand better how to organize and manage project tasks well.
- **The importance of teamwork:** I realized that most IT projects, like cloud migration, usually need a team to be finished properly. It's hard to manage all aspects alone when different roles like network engineers, database admins, and security experts are required.
- **Problem-solving skills:** Because of the limitations of my resources, I had to think creatively and find alternative solutions to meet the project goals under those restrictions.

Technical growth:

- **Cloud computing knowledge:** I knew almost nothing about cloud technologies before this project. Now, I understand the main tools and services in Azure, and I know how to handle cloud migration.

- **Azure cloud configuration:** I also feel more comfortable working with Azure's systems now. I can set up virtual machines, put web applications on the cloud, and manage databases. There are still some advanced tools I didn't get to use much, but this project gave me a good start to learn more.
- **Better MySQL and GitHub Skills:** I improved my use of MySQL and GitHub. I learned how to use GitHub Actions for automation, which helped me complete project tasks more easily.

5.2 Conclusions

Strengths of the project:

- The project demonstrates how a company can use public cloud IT services to move local resources to the cloud, reducing dependence on local physical infrastructure.
- Azure cloud allows flexible resource scaling, on-demand self-service, and real-time cost monitoring. It also enables "pay-as-you-go" based on different periods. (Project cost monitor can be found in Appendix D, Figure 2)

Limitations of the project:

- The student subscription limited access to advanced features and configurations, making some complex cloud services unusable.
- Lack of a real business environment. Without a real company or client as the project base, the configuration and migration felt unnatural and disconnected from actual business needs.
- Time management was a big challenge. I underestimated the complexity of learning and operating cloud services. Two weeks of learning time was far from enough. Also, because the company was virtual, I had to spend extra time creating web apps and data for migration.

Future suggestions:

- With more learning time and resource permissions, I would explore more advanced areas like Kubernetes, Power BI, AI, and more complex security features to improve project quality.
- I recommend practicing cloud migration in a real enterprise environment. This would help understand how cloud systems change with business needs and provide more targeted configurations.
- Cost optimization: In big projects, cost management is very important. We can use the Azure Cost Management tool to monitor and optimize resource usage.

What would I do differently:

- I would give more time to learn at the beginning.
- I would try to migrate more complex datasets, more files, and web applications with more features. Although it would be a larger project, it would bring the project closer to real-world needs.
- I would focus more on cloud environment security early in the project. Solving these problems earlier would make the testing phase smoother.

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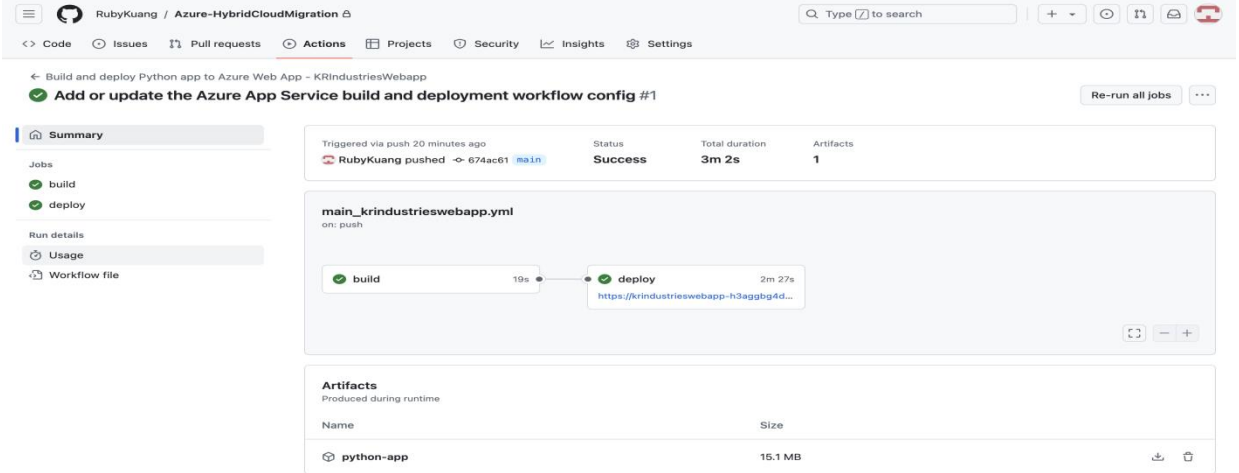
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7. APPENDICES

7.1 Appendix A: Key Configurations and Connections

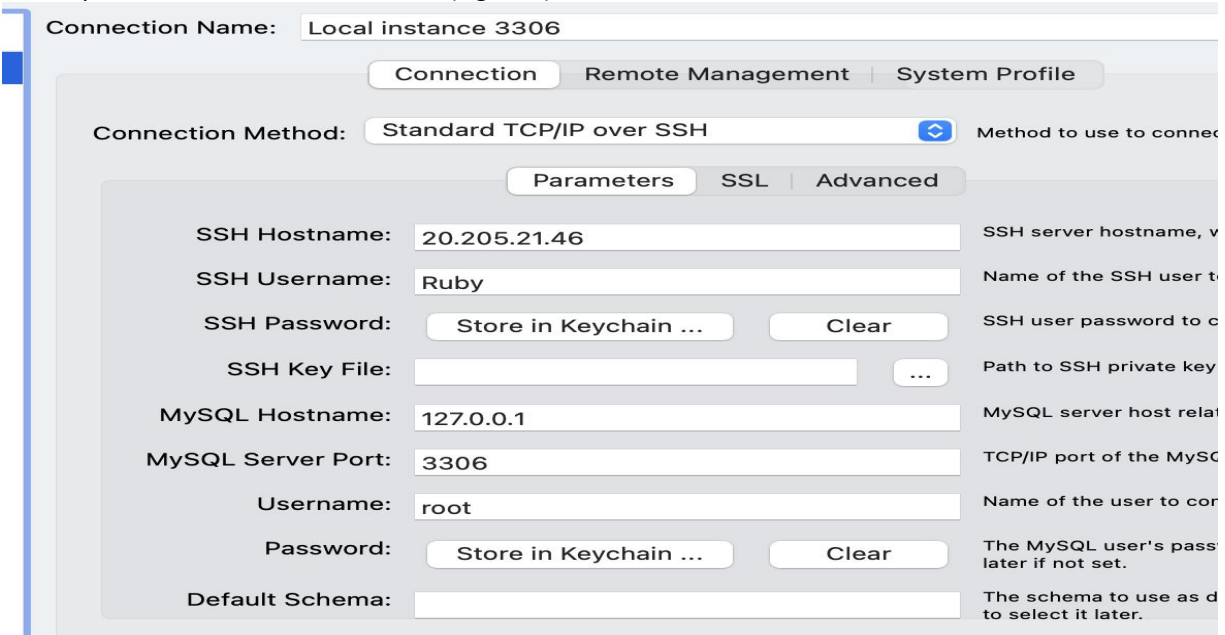
- GitHub Actions Deployment(Figure 1)



<https://github.com/RubyKuang/Azure-HybridCloudMigration/actions/runs/11312560974/usage>

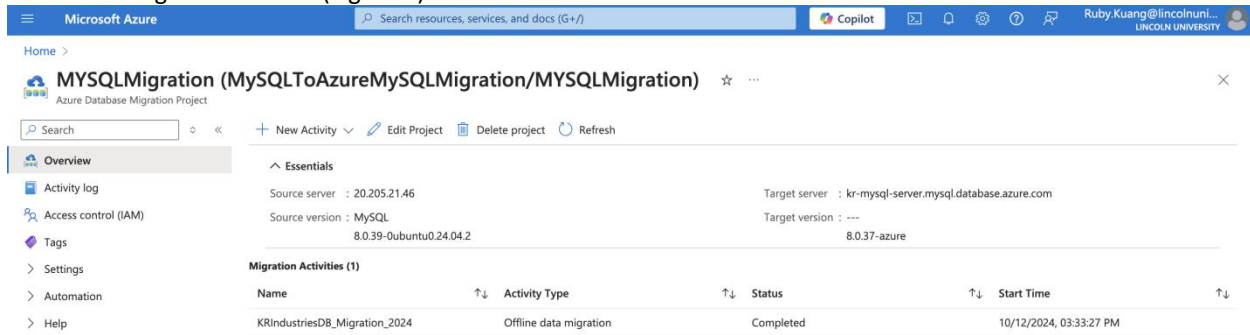
Description: This shows the successful deployment of the web app using GitHub Actions. It confirms the automation of the deployment process.

- MySQL Workbench SSH Connection (Figure 2)



Description: This shows the SSH connection to the Azure MySQL database, which ensures the connection is secure.

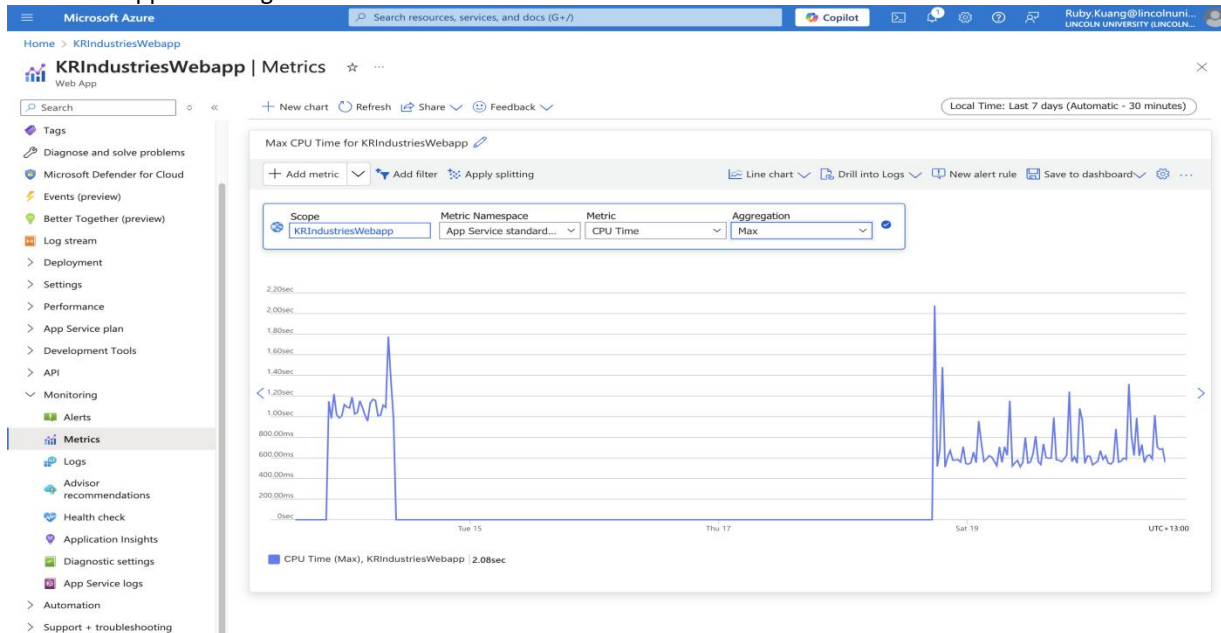
- Data Migration Service (Figure 3)



Description: This screenshot shows the setup and final status of the Data Migration Service (DMS) used to move the MySQL database to Azure. It proves that the offline data migration finished without any issues.

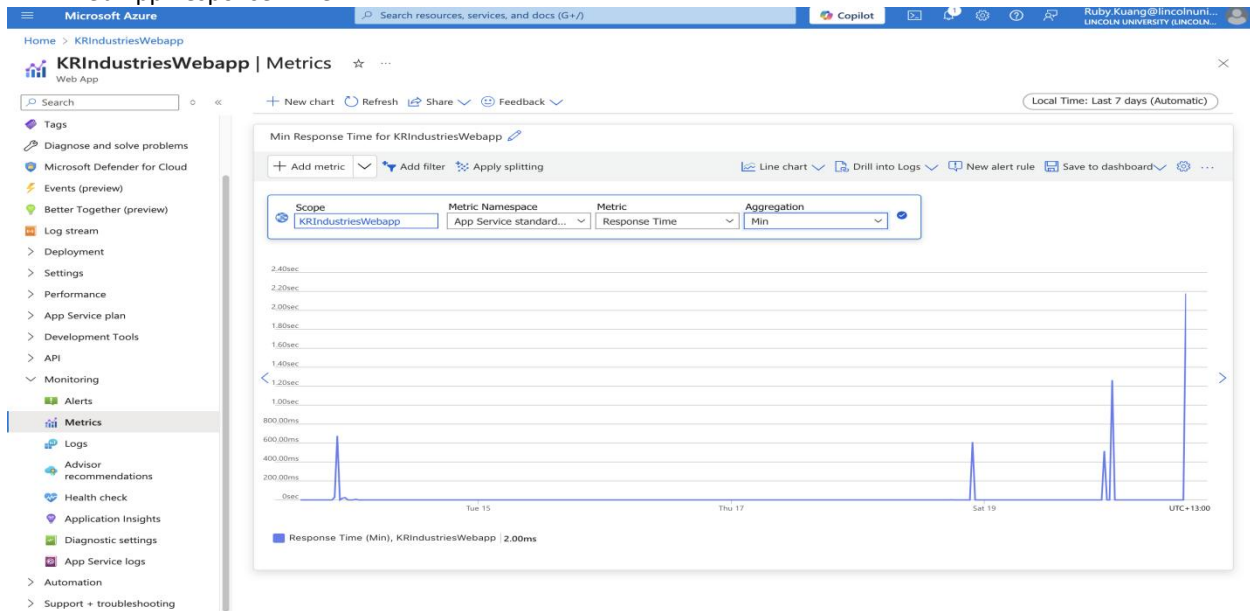
7.2 Appendix B: Performance Metrics

- Web App CPU Usage



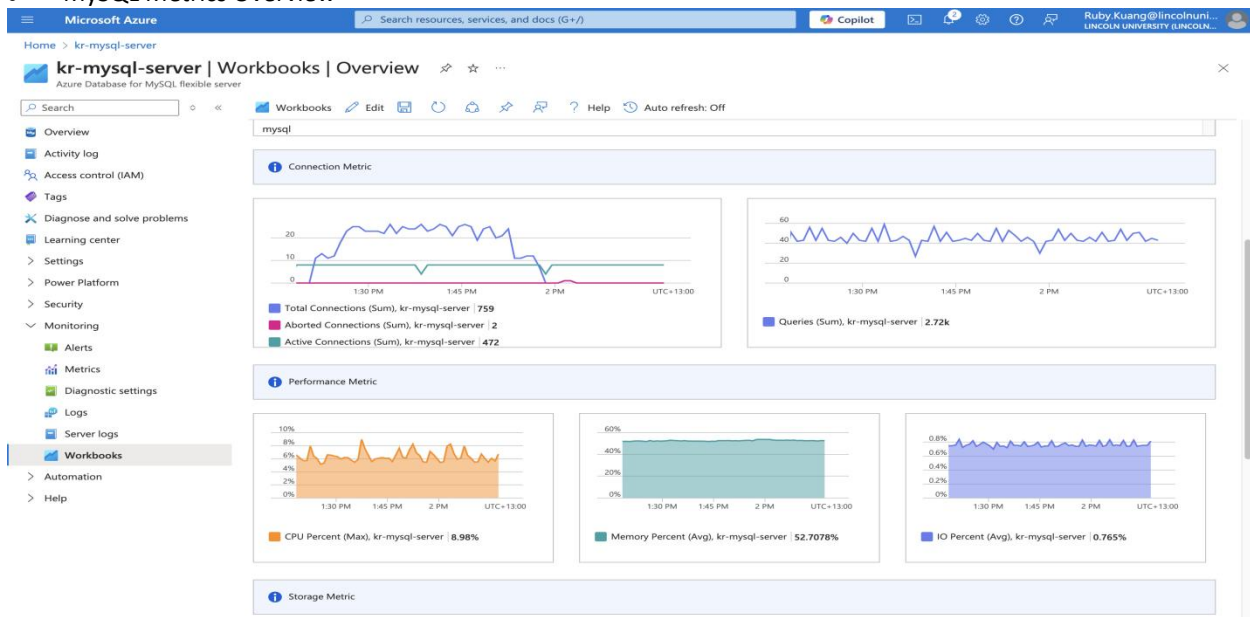
Description: This shows how much CPU is used by the web app. It helps understand the app's performance.

- Web App Response Time



Description: This shows how fast the web app responds. It helps measure the app's speed.

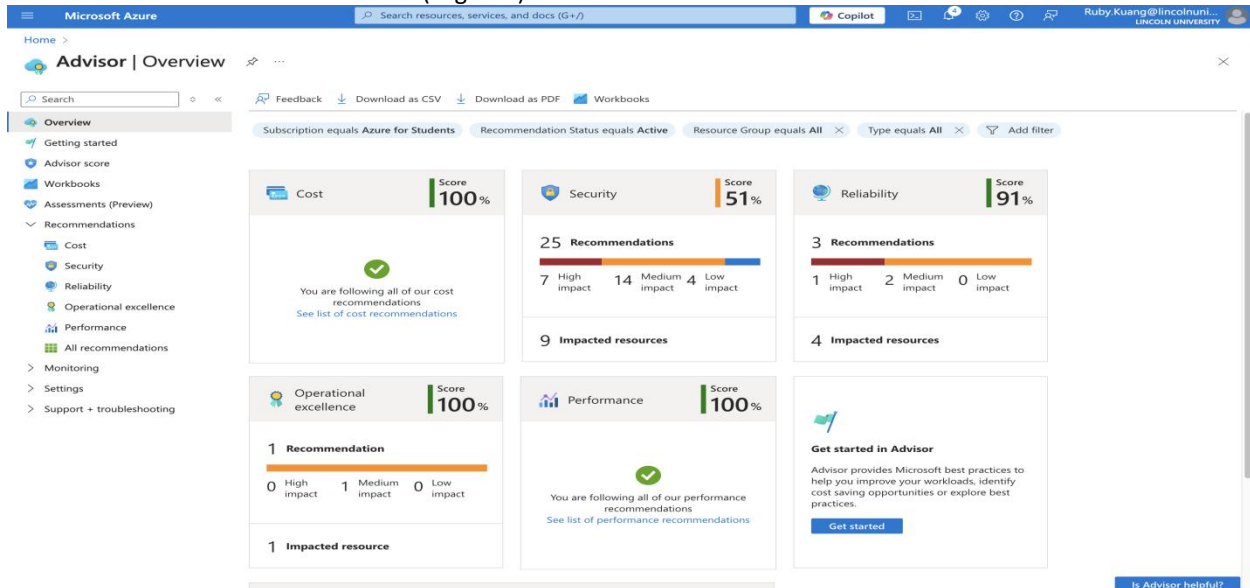
- MySQL Metrics Overview



Description: This shows the performance of the MySQL server, including CPU, memory, and active connections.

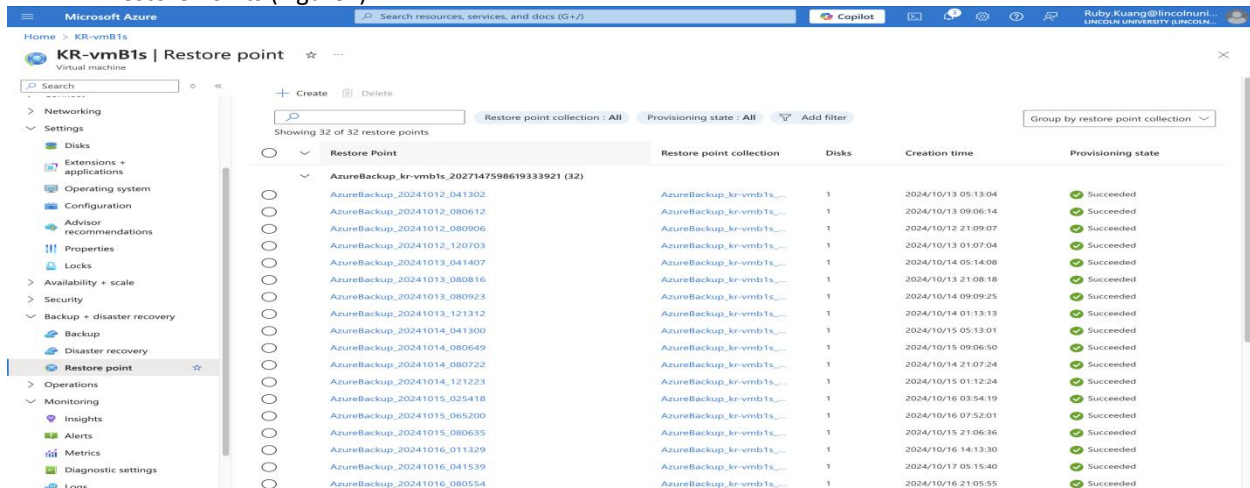
7.3 Appendix C: Security and Backup

- Azure Advisor Recommendations(Figure1)



Description: This shows the Azure Advisor's recommendations for cost, security, and performance. It helps find areas to improve.

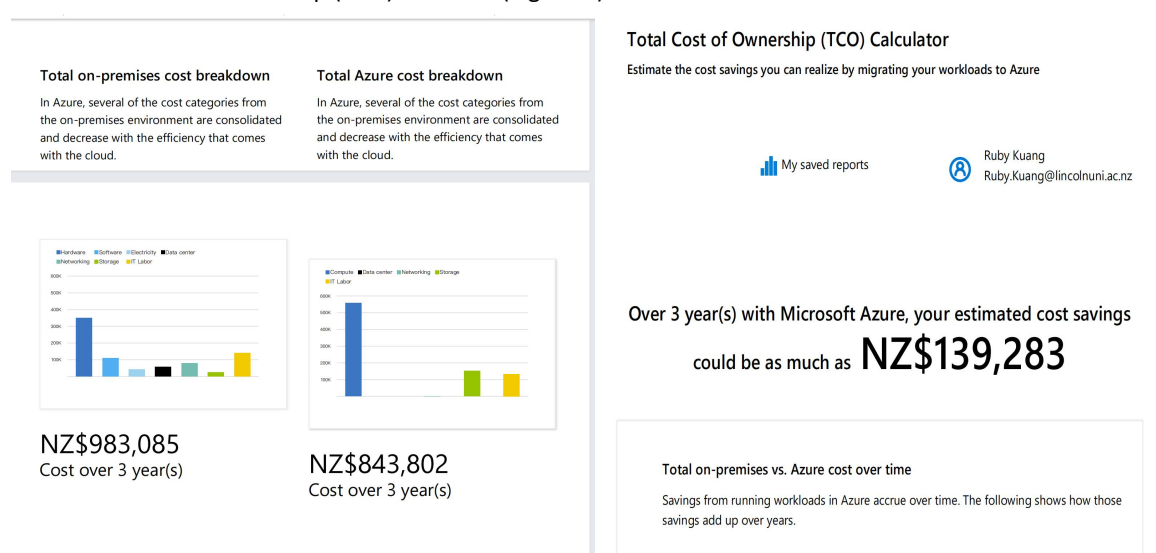
- VM Restore Points (Figure2)



Description: This shows the restore points of the virtual machine. It proves that backup and recovery are working.

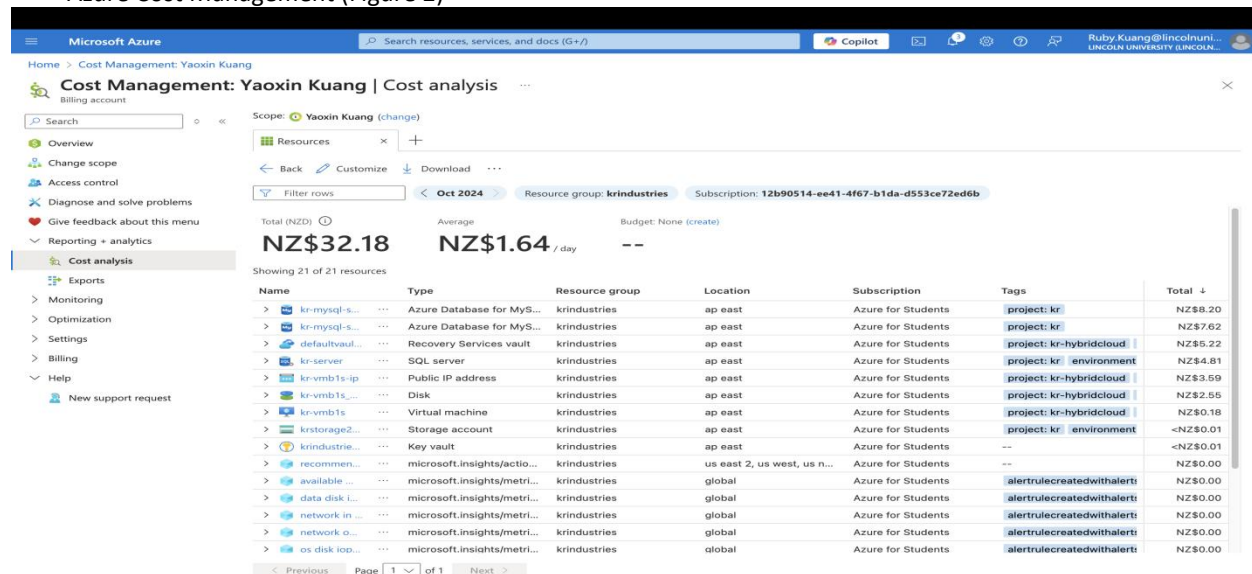
7.4 Appendix D: Cost Analysis

- Total Cost of Ownership (TCO) Estimate (Figure 1)



Description: This summarizes the total on-premises vs. Azure cost breakdown over three years. The estimated savings are NZ\$139,283 based on the initial configurations. For more details on the TCO breakdown, please check in the Github repository folder: <https://github.com/RubyKuang/Azure-HybridCloudMigration/tree/RubyKuang-project-docs>

- Azure Cost Management (Figure 2)



Description: This shows the cost monitoring of the project's resources. It helps track the project's expenses.